

PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
Automated Ocular Disease Classification Using MobileNetV2
Transfer Learning on Fundus Retinal Images
submitted in partial fulfilment of the requirements for the award of degree of

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In
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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Automated Ocular Disease Classification Using MobileNetV2 Transfer Learning on Fundus Retinal Images**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Darpan Anand**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place : Rajpura

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

This paper presents a deep learning-based approach for the detection of ocular diseases using retinal fundus images. Ocular disorders affect a large population worldwide, and early diagnosis remains challenging in regions with limited access to ophthalmic specialists. The proposed system utilizes MobileNetV2- based transfer learning to classify retinal images into four categories: Cataract, Diabetic Retinopathy, Glaucoma, and Normal. The dataset used in this study consists of 4,217 labelled fundus images obtained from a publicly available Kaggle source. The methodology involves data preprocessing, image augmentation, and model training using a convolutional neural network architecture. The model is trained over multiple epochs with early stopping to prevent overfitting. Performance is evaluated using metrics such as accuracy, precision, recall, and loss analysis. Experimental results show that the proposed model achieves a peak validation accuracy of 82.56%, demonstrating effective generalization across multiple disease classes. The findings highlight the potential of lightweight deep learning models for automated eye disease screening and their applicability in resource-constrained environments. This approach can assist in early detection and support clinical decision-making processes.

Keywords— MobileNetV2, Transfer Learning, Eye Disease Classification, Fundus Retinal Imaging, Cataract, Diabetic Retinopathy, Glaucoma, Deep Learning, TensorFlow

1. Introduction

The human eye is among the most intricate sensory organs, enabling perception of the surrounding world through light-mediated neural signaling. Yet this delicate apparatus is vulnerable to a range of progressive disorders that can culminate in irreversible blindness when left undetected. The World Health Organization estimates that over 2.2 billion individuals live with vision impairment globally, and a large fraction of these cases involve conditions that could have been prevented or treated with timely intervention.

Glaucoma silently damages the optic nerve over years, frequently presenting no symptoms until peripheral vision is substantially lost. Diabetic Retinopathy arises from microvascular deterioration in the retinal bed secondary to chronically elevated blood glucose, threatening central vision in an expanding population of diabetic patients worldwide. Cataracts — a progressive opacification of the crystalline lens — remain the foremost cause of correctable blindness and are a disproportionate burden in low-income communities.

Automated fundus image analysis offers a scalable, non-invasive pathway to close this diagnostic gap. Deep learning, in particular Convolutional Neural Networks (CNNs) pre-trained on large image corpora and adapted via Transfer Learning, has achieved ophthalmologist-level performance on retinal grading tasks. MobileNetV2, with its inverted residual architecture and depthwise separable convolutions, offers an especially attractive trade-off: competitive representational power at a fraction of the computational cost of classical deep networks like VGG-16 or ResNet-50.

This project presents a comprehensive implementation of MobileNetV2 for multi-class ocular disease classification, focusing on efficient model design, training, and evaluation across four retinal disease categories using a publicly available dataset of 4,217 fundus images.

1.1 Background of Eye Diseases

A thorough understanding of the target conditions informs the design choices at every stage of the classification pipeline. The three diseases targeted in this study are:

Glaucoma: A cluster of eye disorders that collectively damage the optic nerve — the neural conduit responsible for relaying visual information from the retina to the brain. Elevated intraocular

pressure is the most prevalent risk factor. Damage accumulates insidiously; by the time visual field loss becomes noticeable to the patient, irreversible neuronal loss has already occurred.

Diabetic Retinopathy: An ocular manifestation of systemic diabetes mellitus, arising from prolonged hyperglycemia-induced damage to retinal capillaries. Microaneurysms form initial lesions; progressive ischemia drives hemorrhages, hard exudates, macular edema, and ultimately proliferative neovascularization. It is a leading cause of new-onset blindness in working-age adults.

Cataract: A clouding of the eye's natural crystalline lens that progressively impairs light transmission to the retina, producing blurred or cloudy vision. It remains the single largest contributor to preventable blindness worldwide, amenable to surgical correction if detected early.

2. Methodology

The methodology followed in this project is structured into distinct phases: dataset preparation, data preprocessing and augmentation, model architecture design using Transfer Learning, model compilation and training, and finally evaluation and result analysis.

2.1 Dataset

The Eye Diseases Classification dataset, sourced from Kaggle, organizes retinal fundus images into four class-labelled subdirectories. The dataset contains a total of 4,217 images distributed across four categories as shown in Table 1.

Class	Total Images	Training (80%)	Validation (20%)
Cataract	1,038	830	208
Diabetic Retinopathy	1,098	878	220
Glaucoma	1,007	806	201
Normal	1,074	860	214
Total	4,217	3,374	843

Table 1: Dataset Class Distribution

2.2 Data Preprocessing and Augmentation

All images are resized to 256x256 pixels and pixel values are normalized from [0, 255] to [-1, +1] to match the preprocessing convention of the MobileNetV2 ImageNet pre-training. An 80/20 stratified split is applied using fixed random seeds (seed=123 for training, seed=42 for validation) to ensure reproducibility.

Data augmentation is embedded as a Keras Sequential sub-layer within the model graph, activated only during training. The following transformations are applied:

- Random Horizontal Flip — mirrors the image left-to-right to simulate different viewing angles
- Random Rotation (factor = 0.5) — rotates images up to ± 180 degrees to introduce orientation diversity

Pipeline optimization using `.cache()` and `.prefetch()` with AUTOTUNE is applied to decouple disk I/O from GPU computation, eliminating idle cycles during training.

2.3 Model Architecture — MobileNetV2 with Transfer Learning

Transfer Learning is employed using MobileNetV2 pre-trained on the ImageNet dataset (1.2 million images, 1000 classes). The backbone is loaded with its classification head removed (`include_top=False`) and all 2,257,984 parameters are frozen (`trainable=False`). A custom classification head is added on top:

Layer	Configuration	Output Shape
Input	256x256x3 RGB	256x256x3
RandomFlip	Horizontal	256x256x3
RandomRotation	factor = 0.5	256x256x3
Preprocess Input	Normalize to [-1, +1]	256x256x3

MobileNetV2	2,257,984 params (frozen)	8x8x1280
GlobalAvgPool2D	—	1280
Dropout	rate = 0.20	1280
Dense (output)	4 units, no activation	4

Table 2: Full Model Architecture

2.4 Model Compilation and Training

The model is compiled using the Adam optimizer with a learning rate of 0.0001, and SparseCategoricalCrossentropy loss with from_logits=True. The model is trained for 25 epochs with an EarlyStopping callback (patience=3) monitoring validation loss. In practice, all 25 epochs elapsed, indicating continuous improvement throughout the training run.

3. Tools and Technologies

Tool / Technology	Purpose
Python 3.11	Primary programming language
TensorFlow 2.13 / Keras	Deep learning framework for model building and training
MobileNetV2	Pre-trained CNN backbone for Transfer Learning
NumPy 1.26.4	Numerical array operations and image processing
Matplotlib / Seaborn	Data visualization — accuracy/loss plots
Pillow (PIL)	Image file handling and resizing
OpenCV (cv2)	Image reading and processing
Scikit-learn	Classification metrics (precision, recall, F1-score)
Jupyter Notebook	Interactive development and presentation environment
Kaggle	Dataset source and cloud compute platform

Table 3: Tools and Technologies Used

4. Implementation

The implementation is structured as a step-by-step Jupyter Notebook, documented with markdown explanations at each stage. The following subsections describe each implementation step.

4.1 Library Imports

All required libraries are imported at the outset, including TensorFlow/Keras for model building, NumPy for numerical operations, Matplotlib and Seaborn for visualization, PIL and OpenCV for image handling, and Scikit-learn for evaluation metrics. Warnings are suppressed for a cleaner output.

4.2 Dataset Loading

The dataset is loaded using TensorFlow's `image_dataset_from_directory` utility, which automatically infers class labels from directory names. Images are loaded in batches of 32, resized to 256x256 pixels, and split 80/20 into training and validation sets using fixed random seeds for reproducibility. The four class names — cataract, diabetic_retinopathy, glaucoma, normal — are derived programmatically.

4.3 Data Visualization

A reusable `visualize_images()` function is implemented to randomly sample and display five representative images from each class directory. This step is executed once per class to visually inspect dataset quality and understand the visual characteristics of each disease before training.

4.4 Pipeline Optimization

The `tf.data` pipeline is optimized using `.cache()` to store decoded image tensors in RAM after the first epoch, and `.prefetch(AUTOTUNE)` to queue the next batch while the GPU processes the current one. The training set additionally applies `.shuffle(1000)` to randomize image order each epoch.

4.5 Data Augmentation

Augmentation layers — `RandomFlip` (horizontal) and `RandomRotation` (factor=0.5) — are embedded directly in the model graph as the first layers. This ensures augmentation is applied on-the-fly during training and automatically disabled during validation and inference.

4.6 Model Construction

The MobileNetV2 backbone is loaded with ImageNet weights and its top classification layer removed. All backbone parameters are frozen. A `GlobalAveragePooling2D` layer collapses the 8x8x1280 spatial feature map to a 1280-dimensional vector. `Dropout(0.2)` is applied for regularization, followed by a Dense layer with 4 output units (one per class) producing raw logit scores.

4.7 Model Training

The model is trained using `model.fit()` for 25 epochs with the `EarlyStopping` callback monitoring validation loss (patience=3). Training progresses from 38.6% training accuracy at Epoch 1 to a peak of 83.68% at Epoch 23, with validation accuracy peaking at 82.56% at Epoch 24. The complete 25-epoch training log is recorded in the history object for visualization.

5. Major Findings / Outcomes / Results

5.1 Training Performance

The model was trained for all 25 epochs without early stopping activation, indicating stable and continuous improvement throughout training. Table 4 summarizes the peak performance metrics achieved:

Metric	Value	Epoch
Peak Validation Accuracy	82.56%	24
Peak Training Accuracy	83.68%	23
Minimum Validation Loss	0.4915	22
Minimum Training Loss	0.4488	23
Final Validation Accuracy	81.61%	25
Backbone Parameters	2,257,984 (frozen)	—
Model Size	~8.6 MB	—

Table 4: Peak Performance Metrics

5.2 Accuracy and Loss Analysis

Training and validation accuracy curves rise consistently across all 25 epochs with close mutual tracking. The narrow gap between training accuracy (83.68%) and validation accuracy (82.56%) confirms that the model is not overfitting, and that Dropout(0.2) combined with random augmentation successfully curbs memorization of the training set.

Training loss descends monotonically from 1.3299 at Epoch 1 to a minimum of 0.4488 at Epoch 23, indicating stable gradient flow throughout the training run. Validation loss reaches its minimum of 0.4915 at Epoch 22 before marginally levelling off, signalling that the model capacity was well-matched to the dataset size.

5.3 Prediction Analysis

Per-image inference is demonstrated through a 5x5 prediction confidence grid on one validation batch. Each image is displayed with its ground-truth label, predicted class, and confidence percentage. The majority of predictions are correct across all four classes.

The primary source of misclassification is Glaucoma being incorrectly predicted as Normal. This is attributable to the subtle structural similarity between early-stage glaucomatous and healthy optic

discs — the cup-to-disc ratio differences that distinguish these categories can be as small as 0.05 to 0.10 units, which is within the range of inter-observer variability even for manual graders.

6. Conclusion and Future Scope

6.1 Conclusion

This project successfully demonstrates that MobileNetV2 with Transfer Learning is an effective and computationally efficient approach for automated multi-class ocular disease classification. Operating on 4,217 retinal fundus images across four categories — Cataract, Diabetic Retinopathy, Glaucoma, and Normal — the frozen-backbone configuration trained with Adam ($\text{lr}=0.0001$), Dropout(0.2), EarlyStopping(patience=3), and minimal augmentation achieves a peak validation accuracy of 82.56% at Epoch 24.

The model's lightweight architecture of approximately 8.6 MB at float32 precision establishes MobileNetV2 as a practical candidate for deployment on mobile fundus imaging devices targeting under-served ophthalmic screening programs. The close alignment between training and validation accuracy throughout training confirms effective generalization without elaborate regularization.

These findings highlight the potential of lightweight deep learning models for automated eye disease screening and their applicability in resource-constrained environments where specialist access is limited.

6.2 Future Scope

Several directions extend the current work and represent opportunities for further improvement:

- Fine-tuning the upper layers of the MobileNetV2 backbone — currently frozen entirely — to allow higher-level convolutional filters to adapt to fundus-specific textures, potentially improving accuracy beyond the 82.56% ceiling observed here.

- Incorporating CLAHE (Contrast Limited Adaptive Histogram Equalization) contrast enhancement as a preprocessing step to reduce image quality variance across different acquisition devices.
- Implementing Grad-CAM (Gradient-weighted Class Activation Mapping) to generate visual attention heatmaps overlaid on fundus images, enabling clinician review of the spatial regions driving each prediction.
- Extending the system to severity grading within each disease class — particularly the ETDRS scale for Diabetic Retinopathy — to transform it from a binary screening tool into a clinically actionable grading assistant.
- Post-training int8 quantization via TensorFlow Lite to reduce the model footprint below 3 MB, enabling on-device inference on mobile fundus cameras deployed in community screening programs.

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Appendices

Appendix A: Complete 25-Epoch Training Log

Epoch	Train Acc	Train Loss	Val Acc	Val Loss
1	0.3858	1.3299	0.6572	0.9198
2	0.6175	0.9228	0.7450	0.7639
3	0.6737	0.7868	0.7616	0.6957
4	0.6963	0.7324	0.7722	0.6506
5	0.7519	0.6554	0.7841	0.6196
6	0.7297	0.6427	0.7888	0.5959
7	0.7667	0.5947	0.7841	0.5903
8	0.7597	0.5967	0.7960	0.5675
9	0.7778	0.5596	0.7948	0.5538
10	0.7850	0.5472	0.7817	0.5717
11	0.7709	0.5614	0.8007	0.5448
12	0.7910	0.5248	0.8043	0.5374
13	0.8104	0.5165	0.8055	0.5346
14	0.8005	0.4927	0.8055	0.5303
15	0.8159	0.4871	0.8161	0.5146
16	0.8147	0.4801	0.8090	0.5227
17	0.8049	0.5041	0.8090	0.5155
18	0.8086	0.4836	0.8185	0.5055
19	0.8220	0.4873	0.8209	0.5001
20	0.8129	0.4971	0.8209	0.5019
21	0.8197	0.4692	0.8114	0.5055

22	0.8247	0.4541	0.8233	0.4915
23	0.8368	0.4488	0.8233	0.4938
24	0.8202	0.4630	0.8256	0.4943
25	0.8227	0.4704	0.8161	0.4990

Table A1: Complete 25-Epoch Training Log